

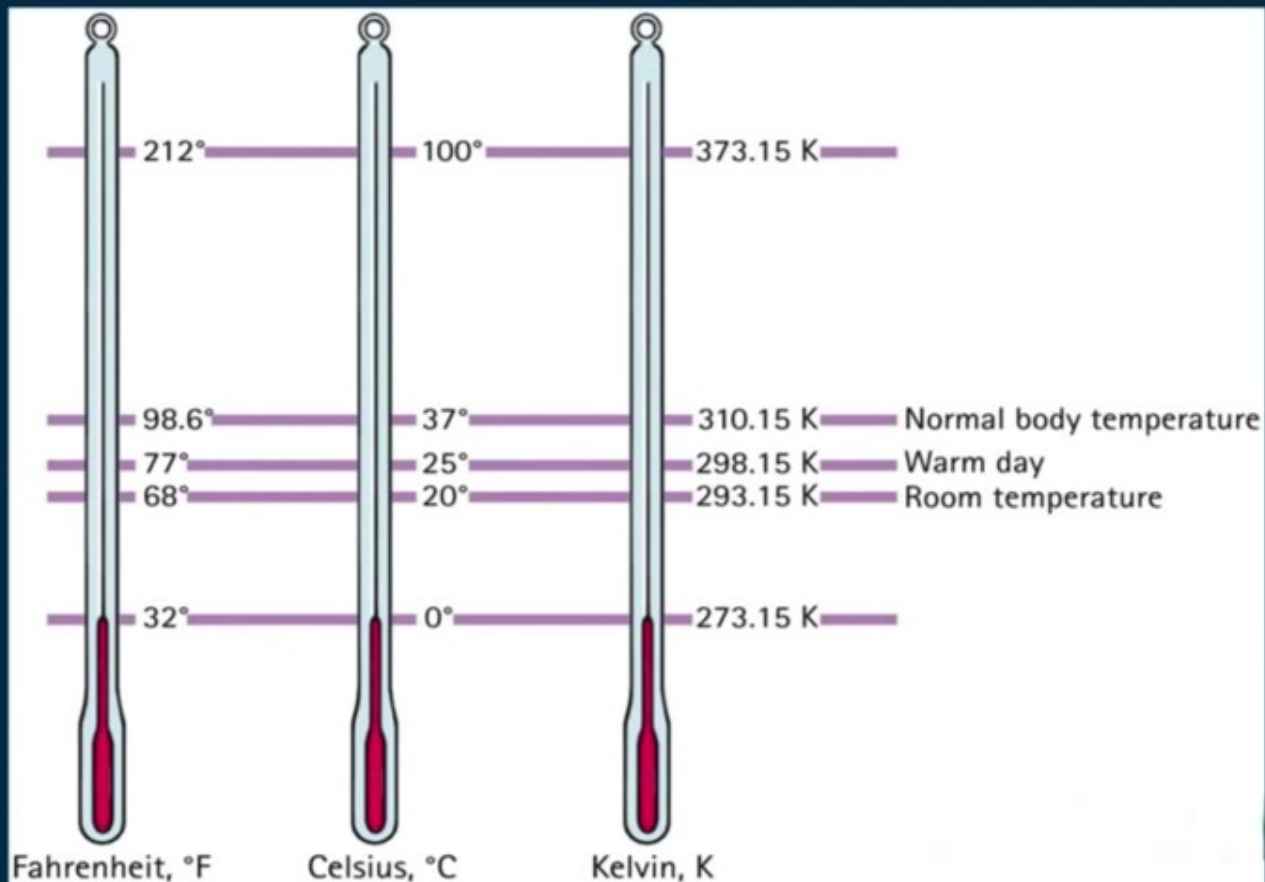
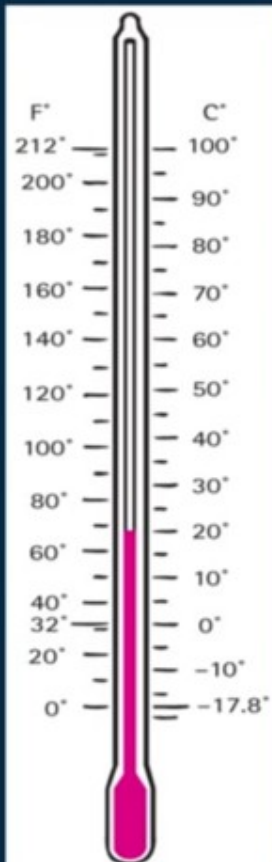
Temperature, Heat Q

heat capacity C , specific heat c

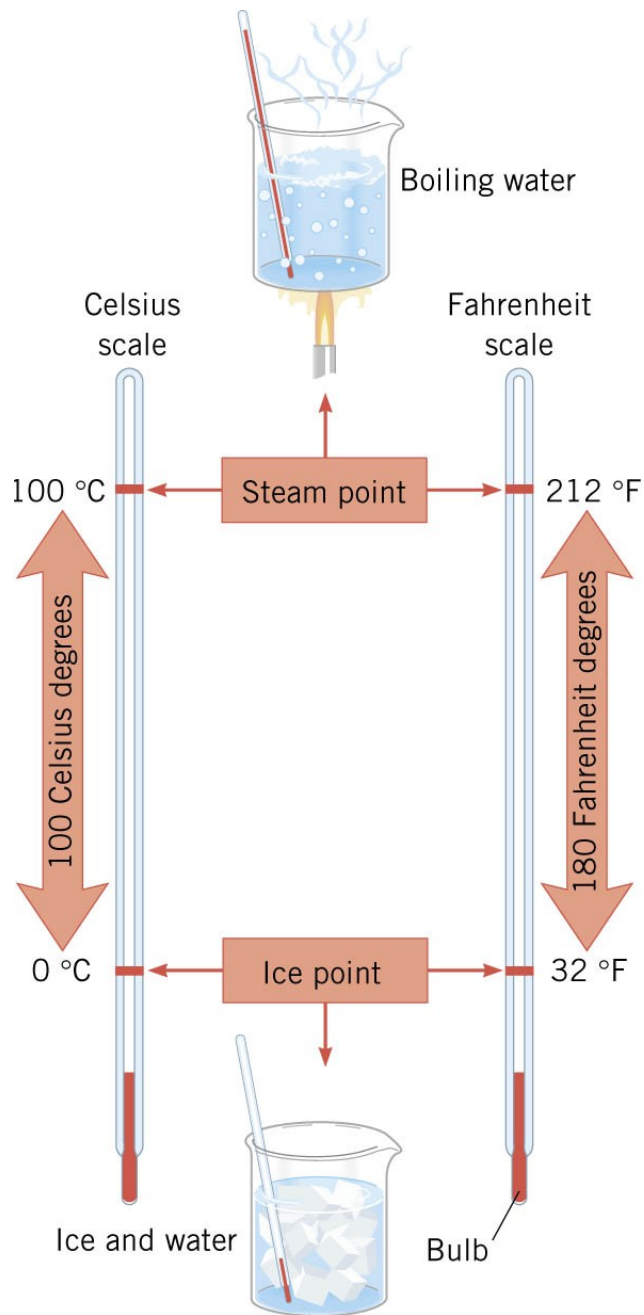
References: - Physics / Johnson and Cutnell / Wiley (slides)
- Conceptual Physics / Paul Hewitt / Pearson
- Physics for future presidents / Dr. Richard Muller / Princeton



- Conceptual Physics / Paul Hewitt / Pearson



12.1 Common Temperature Scales



Temperatures are reported in degrees **Celsius** or degrees **Fahrenheit**.

Temperatures changed, on the other hand, are reported in **Celsius** degrees or **Fahrenheit** degrees:

$$1\text{ }^{\circ}\text{C} = \frac{9}{5}\text{ }^{\circ}\text{F}$$

12.1 Common Temperature Scales

Example 1 Converting from a Fahrenheit to a Celsius Temperature

A healthy person has an oral temperature of 98.6°F . What would this reading be on the Celsius scale?



12.1 Common Temperature Scales

Example 1 Converting from a Fahrenheit to a Celsius Temperature

A healthy person has an oral temperature of 98.6°F . What would this reading be on the Celsius scale?

$$\begin{aligned} 98.6^{\circ}\text{F} - 32^{\circ}\text{F} &= 66.6^{\circ}\text{F} \\ \Downarrow \\ (66.6^{\circ}\text{F}) \left(\frac{1^{\circ}\text{C}}{\frac{9}{5}^{\circ}\text{F}} \right) &= 37.0^{\circ}\text{C} \\ \Downarrow \\ 0^{\circ}\text{C} + 37.0^{\circ}\text{C} &= 37.0^{\circ}\text{C} \end{aligned}$$

degrees above ice point

ice point

12.1 Common Temperature Scales

***Example 2* Converting from a Celsius to a Fahrenheit Temperature**

A time and temperature sign on a bank indicates that the outdoor temperature is -20.0°C . Find the corresponding temperature on the Fahrenheit scale.



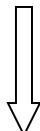
12.1 Common Temperature Scales

Example 2 Converting from a Celsius to a Fahrenheit Temperature

A time and temperature sign on a bank indicates that the outdoor temperature is -20.0°C . Find the corresponding temperature on the Fahrenheit scale.

$$(20.0\text{ }^{\circ}\text{C})\left(\frac{\frac{9}{5}\text{ }^{\circ}\text{F}}{1\text{ }^{\circ}\text{C}}\right)=36.0\text{ }^{\circ}\text{F}$$


degrees below ice point

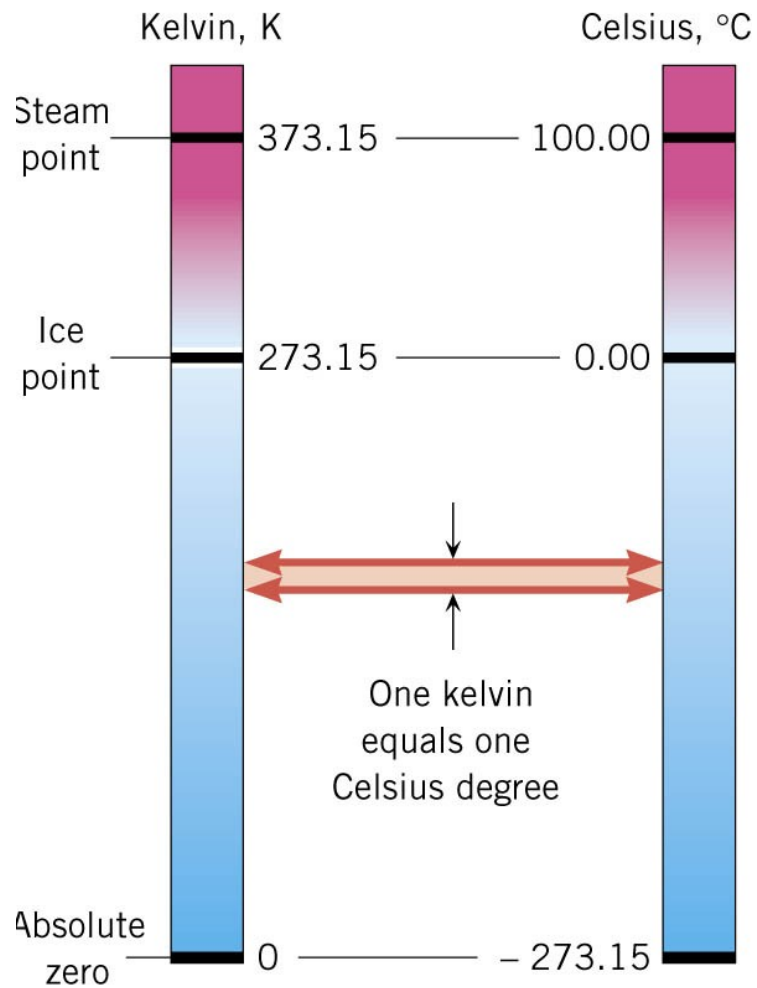


$$32.0\text{ }^{\circ}\text{F}-36.0\text{ }^{\circ}\text{F}=-4.0\text{ }^{\circ}\text{F}$$

ice point



12.2 The Kelvin Temperature Scale



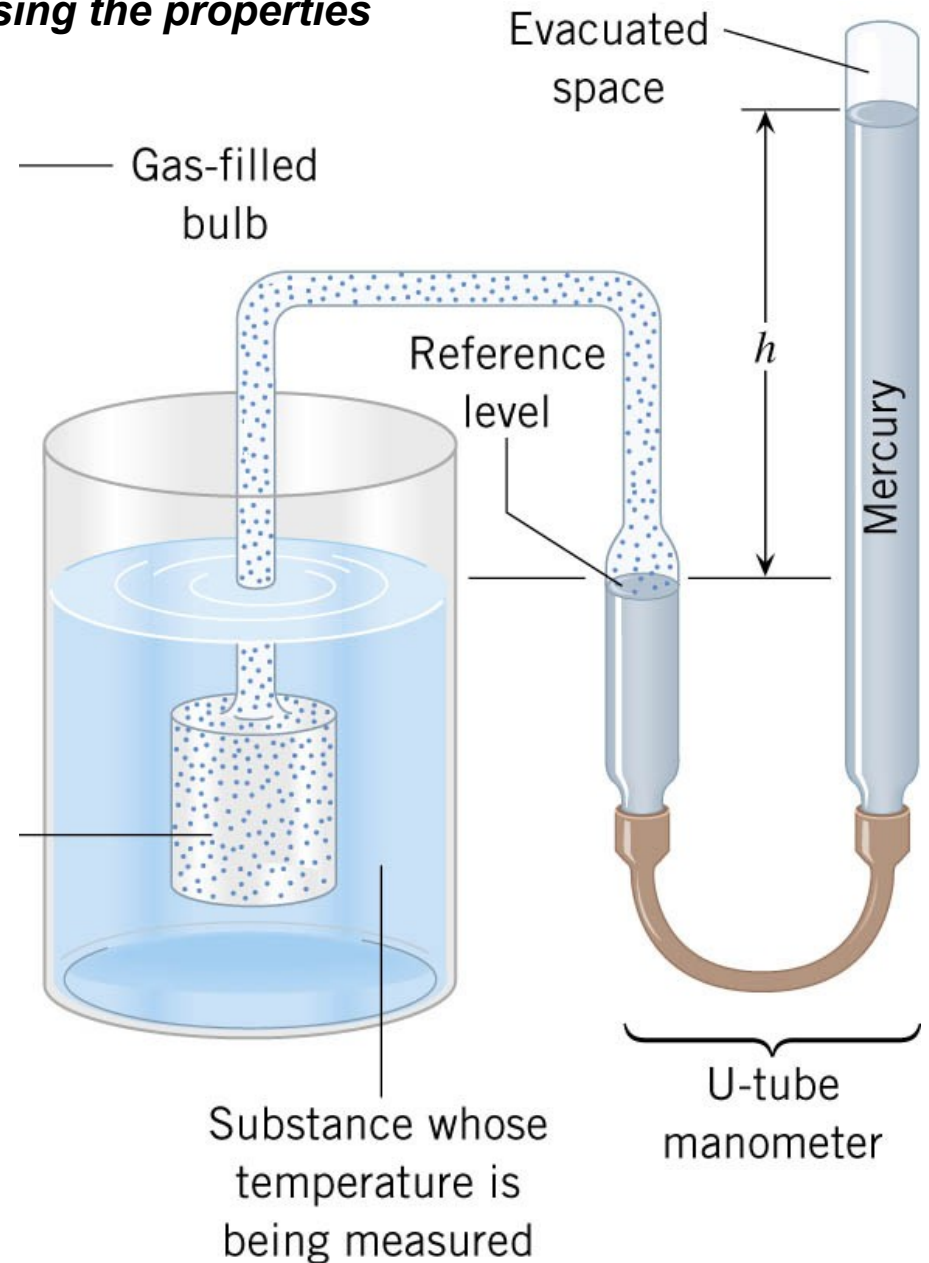
Kelvin temperature

$$T = T_c + 273.15$$

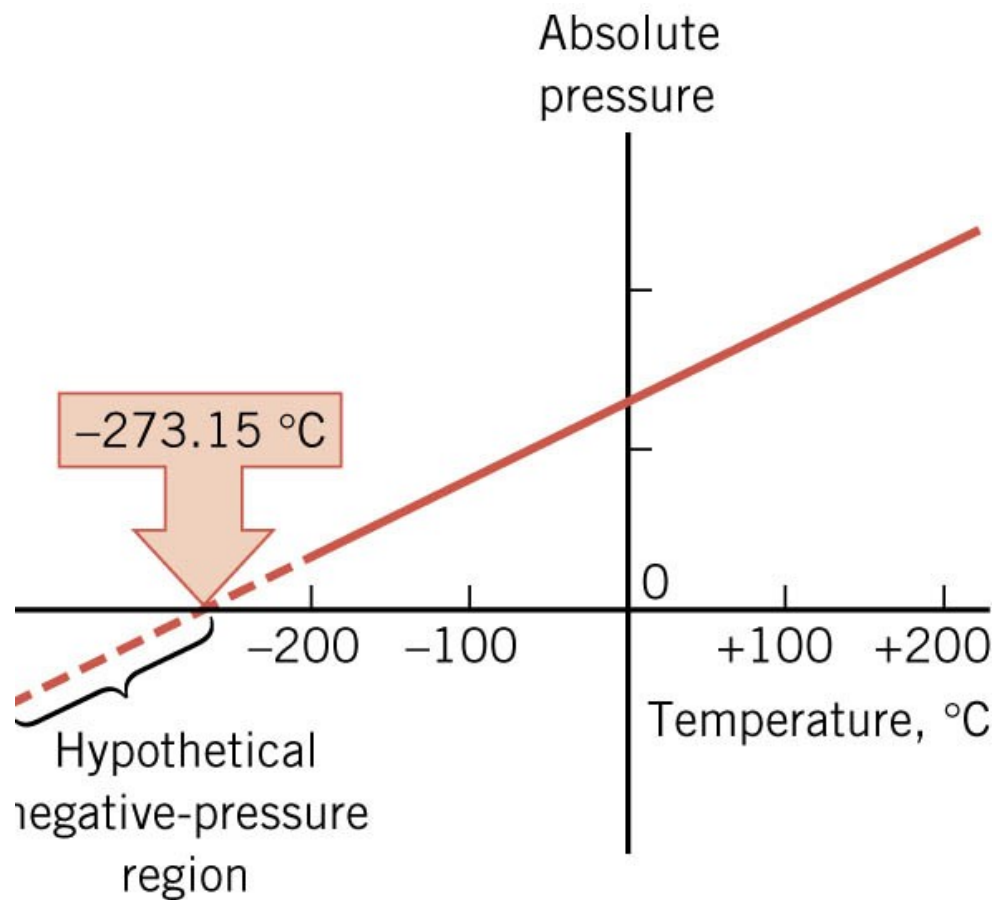
Different ways to measure temperature using the properties of materials.

This method relies on a property of gases known as the ideal gas law.

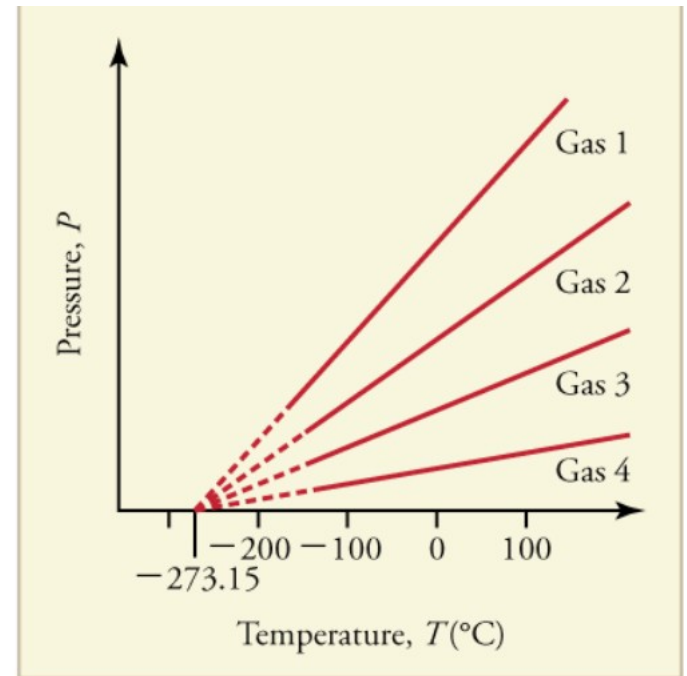
As the expansion of gas is large, so the Gas thermometer are very sensitive. The gas expand uniformly and regularly over a wide range of temperature. The expansion coefficient of all gases is nearly the same, so thermometer using different gases give same reading



12.2 The Kelvin Temperature Scale

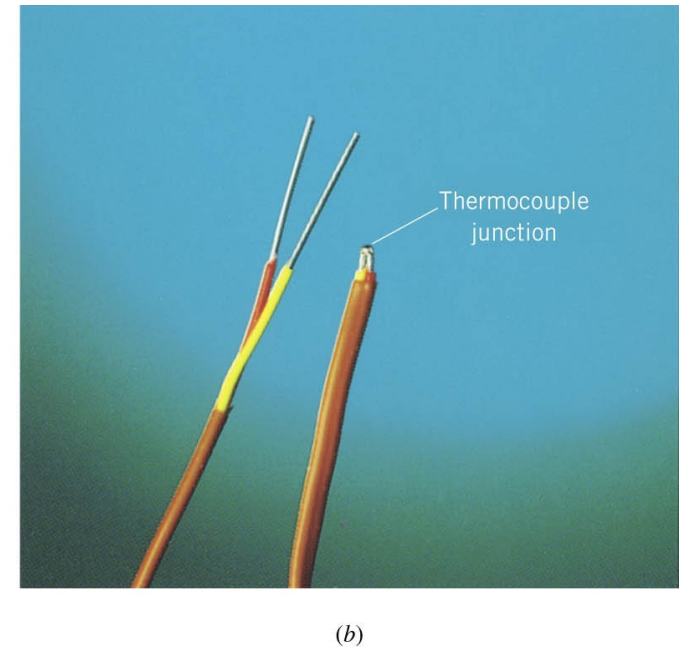
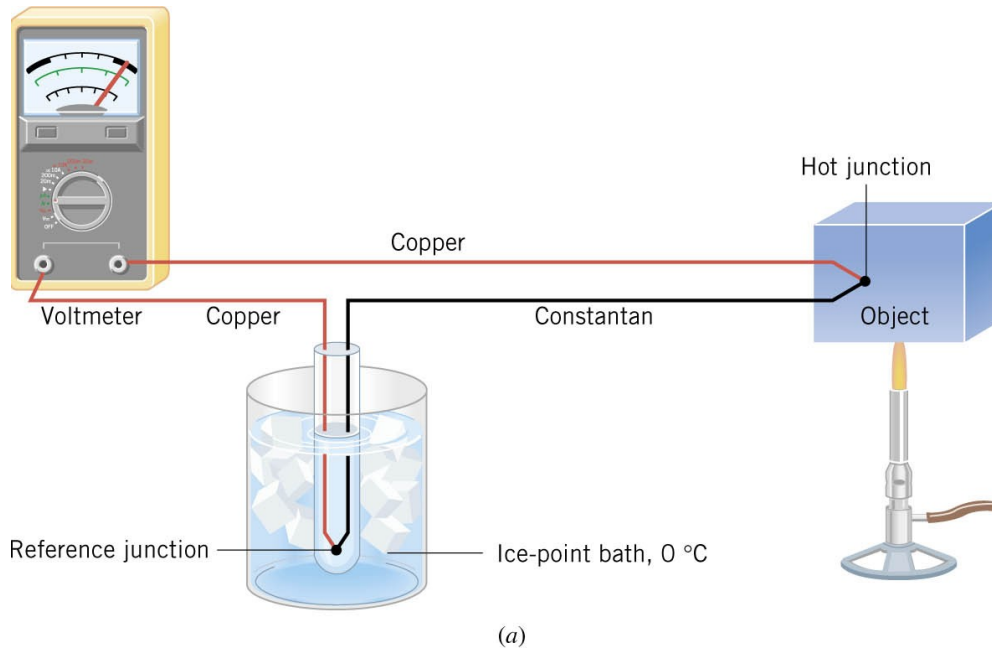


absolute zero point = -273.15°C

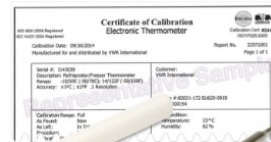
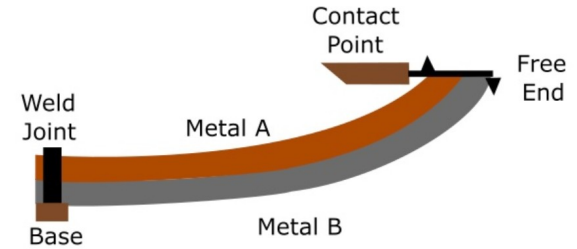
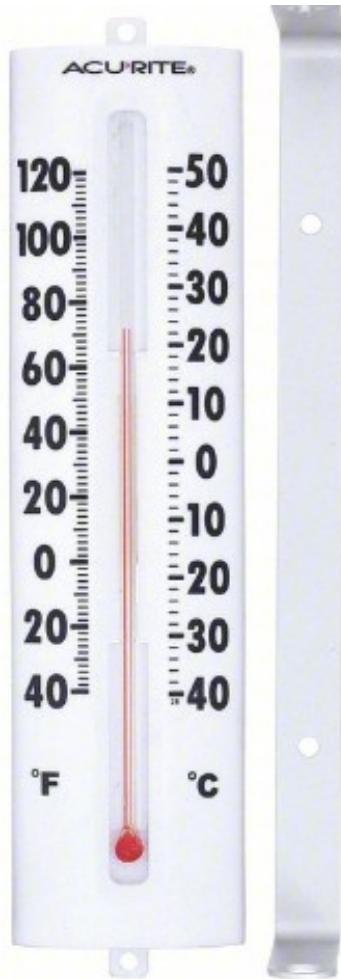


12.3 Thermometers, transducers

Thermometers make use of the change in some physical property with temperature. A property that changes with temperature is called a **thermometric property**.



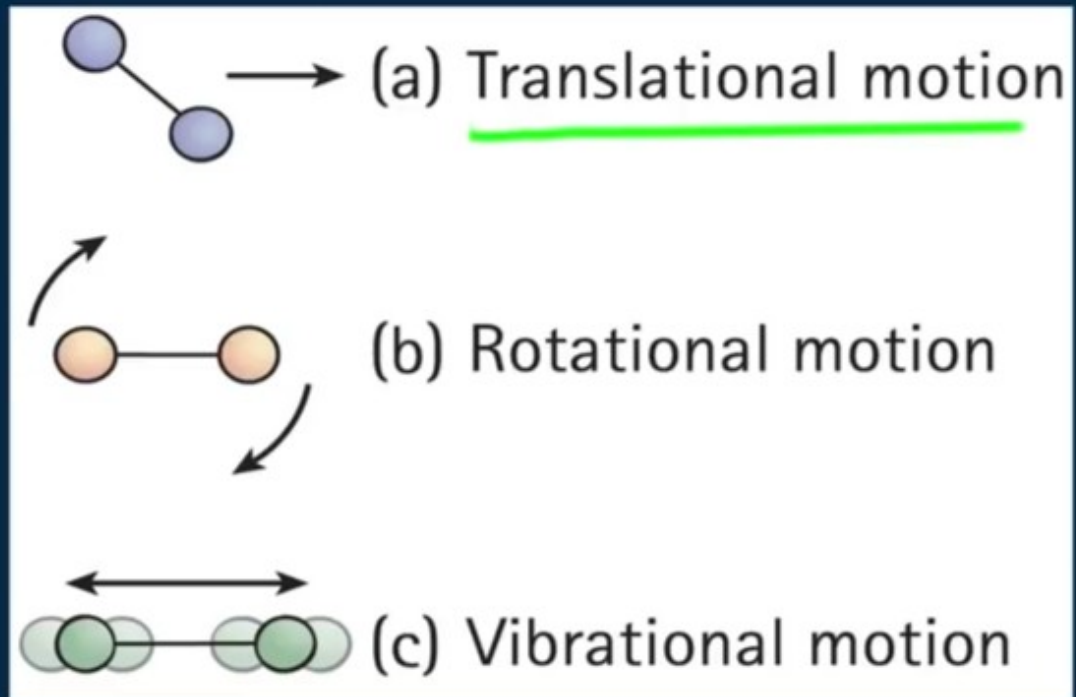
A thermocouple is a device for measuring temperature. It comprises two dissimilar metallic wires joined together to form a junction. When the junction is heated or cooled, a small voltage is generated in the electrical circuit of the thermocouple which can be measured, and this corresponds to temperature.



gif

<https://www.tec-science.com/thermodynamics/temperature/how-does-a-bimetallic-strip-thermometer-work/>

TEMPERATURE



Temperature

- Temperature is proportional to the average translational kinetic energy per particle in a substance.
 - Gas—how fast the gas particles are bouncing to and fro
 - Liquid—how fast particles slide and jiggle past one another
 - Solid—how fast particles move as they vibrate and jiggle in place

To say that body A has a higher temperature than body B is to say that body A has more

- A. internal energy.
- B. mass.
- C. kinetic energy per particle.
- D. potential energy.

TEMPERATURE MEASURES the kinetic energy motion of molecules = the energy of their RANDOM MOTION

**SAME TEMPERATURE =
SAME AVERAGE KINETIC ENERGY per molecule**
 $\frac{1}{2} \times \text{mass} \times \text{speed}^2$

The average kinetic energy per molecule is given by the formula

$$\text{KE} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

T is the temperature in kelvin

KE is the kinetic energy in joules.

TEMPERATURE MEASURES THE RANDOM MOTION OF MOLECULE

Molecules Of different masses have the Same T (KE) but not speed !!

HOW FAST MOLECULES MOVE?

At the speed of Sound !!

About 1 mile in 5 second, 1000 ft in 1 s,
340m/s at room temperature !! (343m/s)

Do you think the sound move faster in solid or gas?
At high temperature or low?

Speed of sound = Mach 1

$T = 300 M^2$ so for $M=1$, $T = \dots$ K or 80 F (27C)
 $M=2$, $T = \dots$ F

If you double the speed , temperature is X 4 →
Explain why must meteors vaporizes/burns up
before hitting the Earth (rub your hands together. You force the
molecules into motion → temperature increases)

Temperature is the average transnational kinetic energy of the molecules
In a substance. The motion of the molecules are randon.

If too much jiggle-ling = bonds break , solid to gas and the system expands
By a factor of 1000 for each increase of 1 C.

A shuttle in orbit around the earth has a speed of 7,000m/s (16000mph)
The speed of the sound is 340 m/s. So the shuttle moves at 20 times the speed
of sound

But no problem because most of the KINETIC ENERGY of the molecules
Of the shuttle is in the the forward motion of the shuttle. Not in their random
motion.

What happens when the shuttle enters the atmosphere ?
From the point of you of the shuttle, the air molecules hit it at a speed
Of about _____ times the molecule of sound. It slows down but the random
Motion of the molecules increases to match the motion of air molecules.

The speed of molecules increases by a factor of 20 !!!

$$T = 300 M^2$$

So the temperature reaches _____ C !!!
The surface of the Sun is 6000 C, what is the factor ?

What happened to Columbia in 2003 ?





Forward kinetic energy of a asteroid = 165 x its own mass of TNT
The kinetic energy goes into random motion of molecules in the asteroid while moving in the atmosphere = heat when it reaches the Earth.

The asteroid that killed the dinosaurs =
 2.6×10^8 megatons of TNT = 200 million megatons of TNT.

What is heat? Some kind of fluid as thought by Lavoisier, the father of modern chemistry

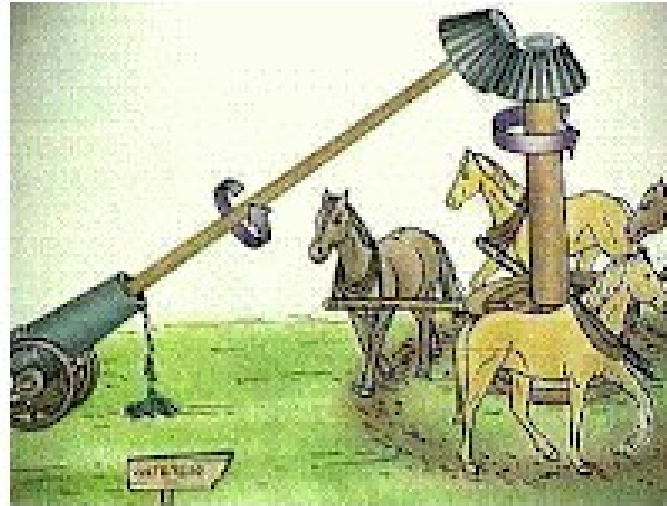
Caloric Theory of Heat: Heat is an invisible fluid called the “caloric” that flows between objects with different temperatures. The caloric naturally flowed from a hot object with a lot of it, to a cold object with only a little of it. The cold object expanded as it warmed because it had absorbed the invisible fluid.





Benjamin Thompson

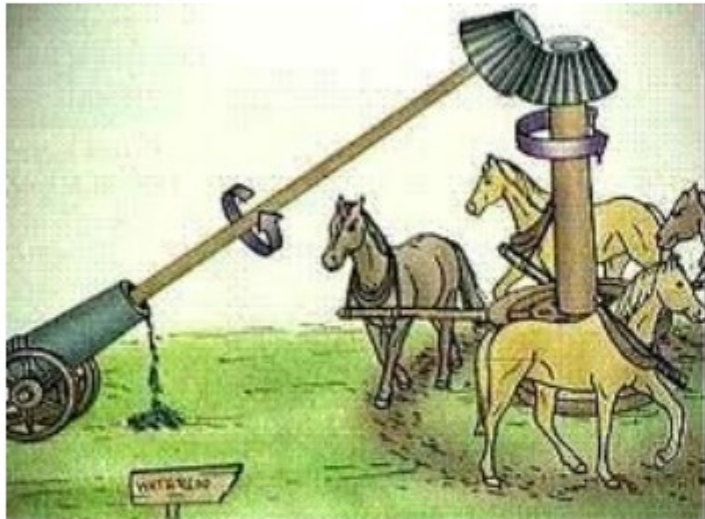
1753 – 1814



Count Rumford's cannon experiment

Rumford had observed the frictional heat generated by boring out cannon barrels at the arsenal in Munich.

Lord Rumford was boring canons (for German army) and came to the conclusion that heat was not a substance. As long as you keep boring, the canon was getting hot. (move your hands together, they get also hot).

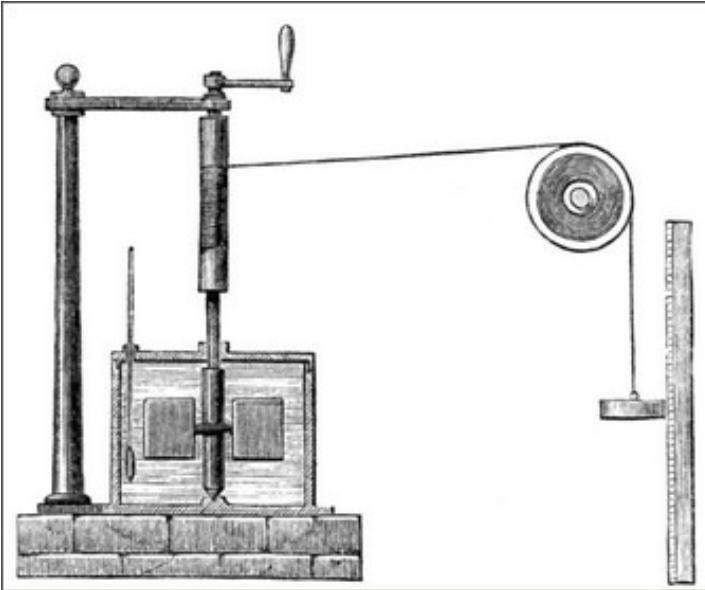


(Benjamin Thomson / Count Rumford)

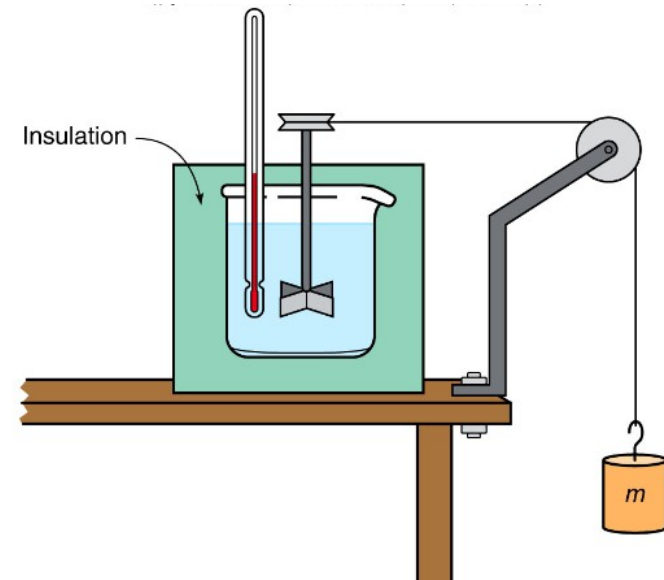
His idea was later developed by James Prescott Joule.
Thermodynamics is the branch of Physics that study heat.
See simulation 1st law from exploration Physics
(1st law = energy conservation)

heat is simply the transfer of energy from a hot object to a colder object.

1850 → James Joule shows that thermal energy is a form of energy/ You can Turn potential energy (mechanical energy) to thermal energy → increase in temperature He quantified the relationship between joules and calories . He found that: it takes about 4 J (1 cal) to increase the temperature of 1 g water by 1 degree. Or about 4000 J (1Kcal) to increaseThe temperature of 1 kg of water by 1 degree.



Joule's apparatus for measuring the mechanical equivalent of heat. A descending weight attached to a string causes a paddle immersed in water to rotate and the "work" of the falling weight is converted into "heat" by agitating the water and raising its temperature.



Applet : Exploration of physics/
1st law of thermodynamics

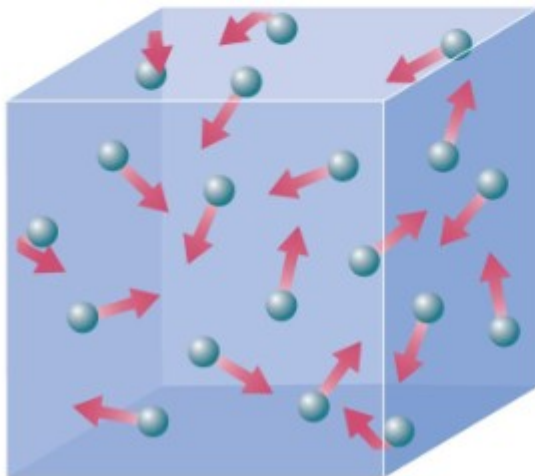
Thermal Energy:

the collective kinetic energy of many particles
(for example, in a rock, in air, in water)

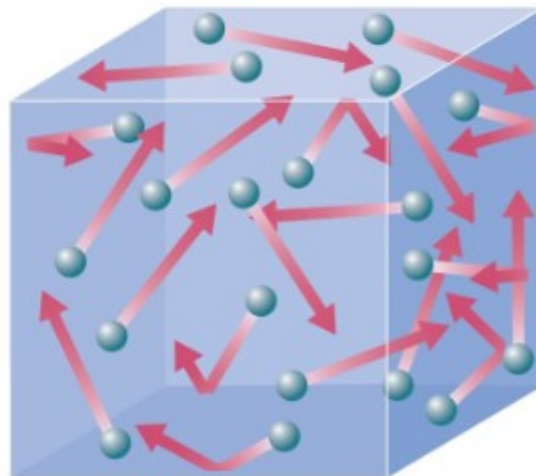
Thermal energy is related to temperature but it is NOT the same.

Temperature is the *average* kinetic energy of the many particles in a substance.

lower temperature



higher temperature

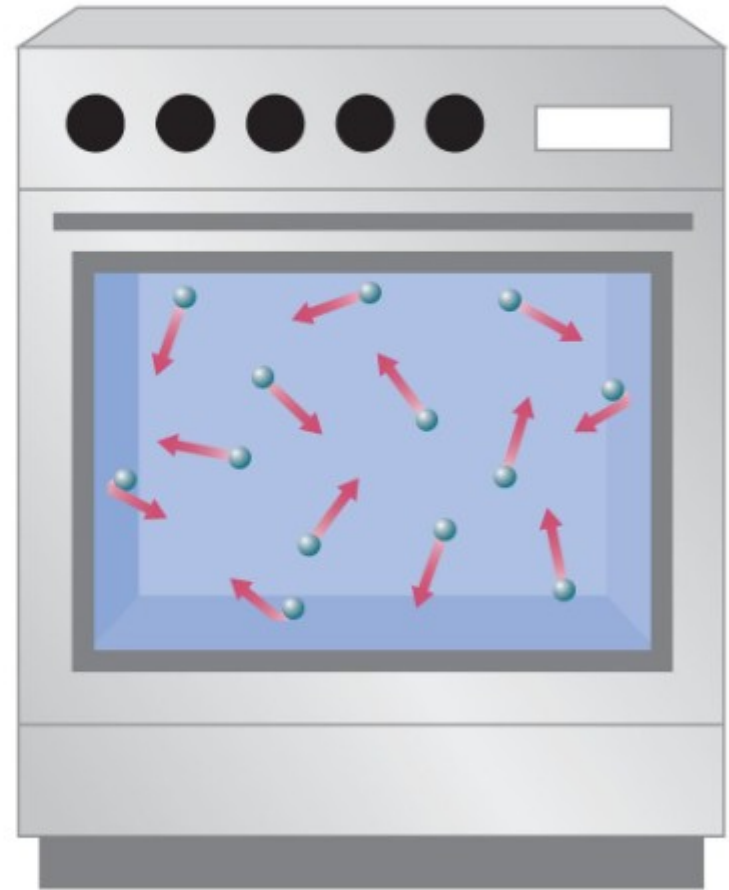


Thermal energy is a measure of the total kinetic energy of all the particles in a substance. It therefore depends on both *temperature AND density*.

Example:



212°F



400°F



Sparkler sticks have a high temperature (3000 C) so large KE / molecule
But few molecules → total amount of Thermal Energy (all molecules) is small.

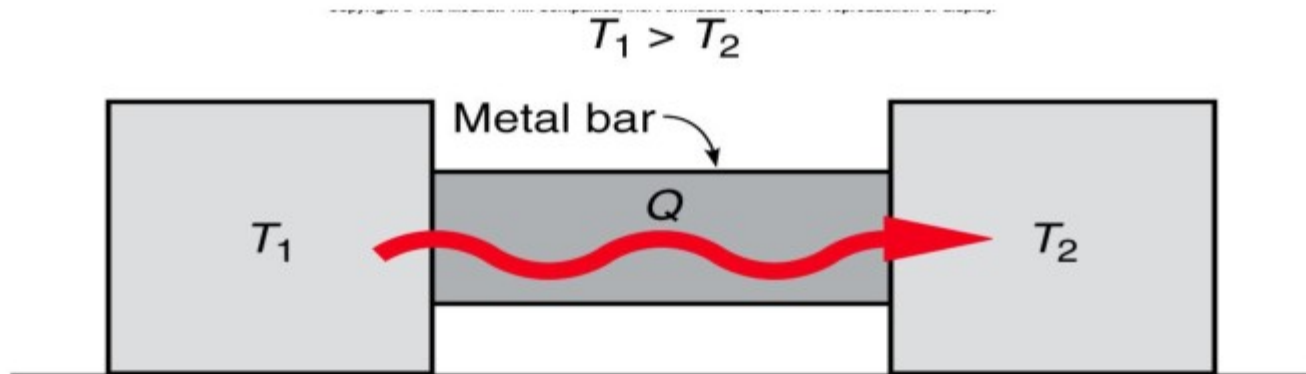


If you walk fast, not a lot of heat can flow.
Plus air is not a good conductor

HEAT = transfer of energy from hot to cold !

Molecules are losing kinetic energy by bumping into slower one.
In a close system all objects tend to the same temperature.
(if no source of energy)

If all objects are at the same temperature in a room why metal seem cooler that other objects when we touch them ?



vapor/air is a bad heat conductor

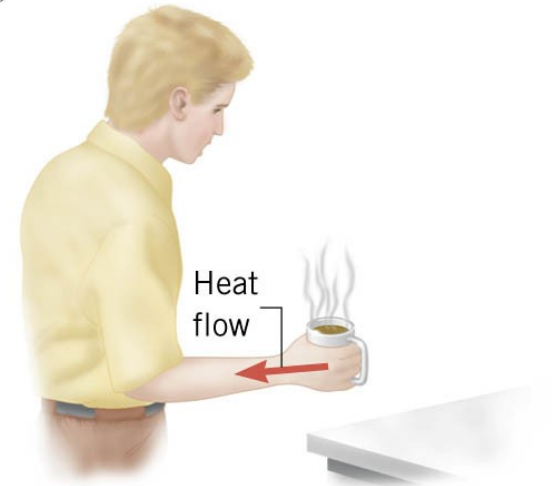
12.6 Heat and Internal Energy

DEFINITION OF HEAT

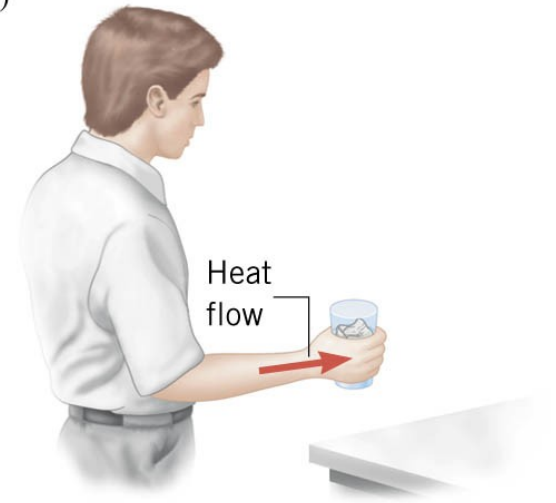
Heat is energy that flows from a higher-temperature object to a lower-temperature object because of a difference in temperatures.

SI Unit of Heat: joule (J)

(a)



(b)



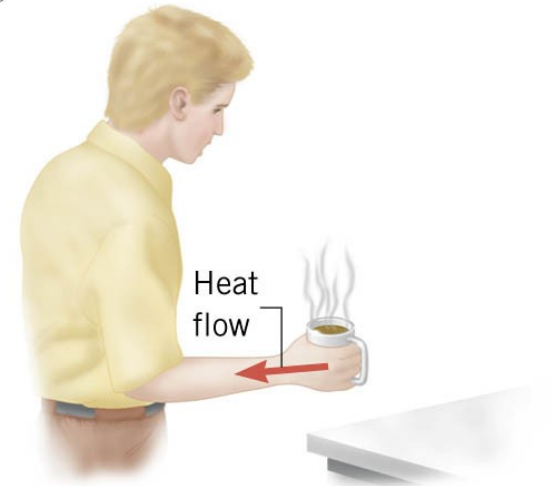
12.6 Heat and Internal Energy

The heat that flows from hot to cold originates in the *internal energy* of the hot substance.

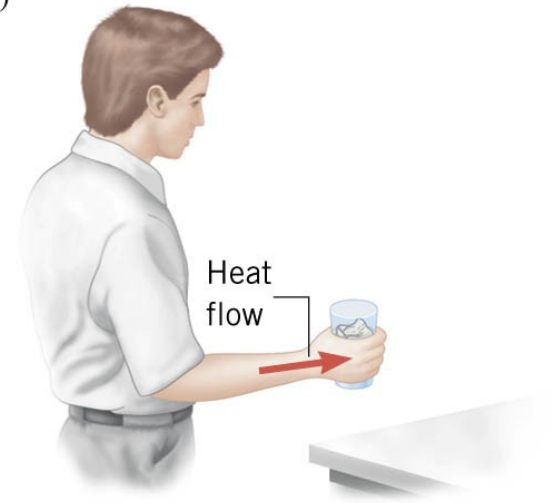
It is not correct to say that a substance contains heat.



(a)



(b)



Quantity of Heat

- Quantity of heat
 - Measured in joules or calories
 - 4.18 joules of heat are required to change the temperature of 1 gram of water by 1 Celsius degree
 - 4.18 joules = 1 calorie
- 1btu is the quantity of heat required to raise the temperature of 1 pound of Water by 1 F.

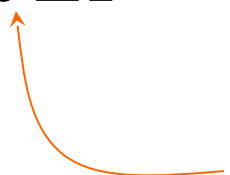
12.7 Heat and Temperature Change: Specific Heat Capacity

SOLIDS AND LIQUIDS

HEAT SUPPLIED OR REMOVED IN CHANGING THE TEMPERATURE OF A SUBSTANCE

The heat that must be supplied or removed to change the temperature of a substance is

$$Q = mc \Delta T$$



specific heat
capacity

Common Unit for Specific Heat Capacity: $\text{J}/(\text{kg} \cdot \text{C}^\circ)$

c is a property of a material. It's the specific heat.

$C = mc$ is the property of an object. It's the heat capacity.

12.7 Heat and Temperature Change: Specific Heat Capacity

Table 12.2 Specific Heat Capacities^a
of Some Solids and Liquids

Substance	Specific Heat Capacity, c $\text{J}/(\text{kg} \cdot ^\circ\text{C})$
Solids	
Aluminum	9.00×10^2
Copper	387
Glass	840
Human body (37 °C, average)	3500
Ice (−15 °C)	2.00×10^3
Iron or steel	452
Lead	128
Silver	235
Liquids	
Benzene	1740
Ethyl alcohol	2450
Glycerin	2410
Mercury	139
Water (15 °C)	4186

^aExcept as noted, the values are for 25 °C and 1 atm of pressure.

- Specific heat capacity
 - Defined as the quantity of heat required to change the temperature of a unit mass of the substance by 1 degree Celsius
 - Like thermal inertia—resistance of a substance to a change in temperature

Material	$c, \text{J}/\text{kg} \cdot \text{K}$	$c, \text{cal}/\text{gm} \cdot ^\circ\text{C}$
Aluminum	900	0.215
Glass	750	0.18
Iron	447	0.107
Stone (granite)	840	0.2
Water, liquid	4184	1.00
Water, ice	2050	0.49
Wood	1400	0.33

For a given mass. It takes 20 min to bring water
To boiling temperature. 2 min for iron. 1 min for silver.

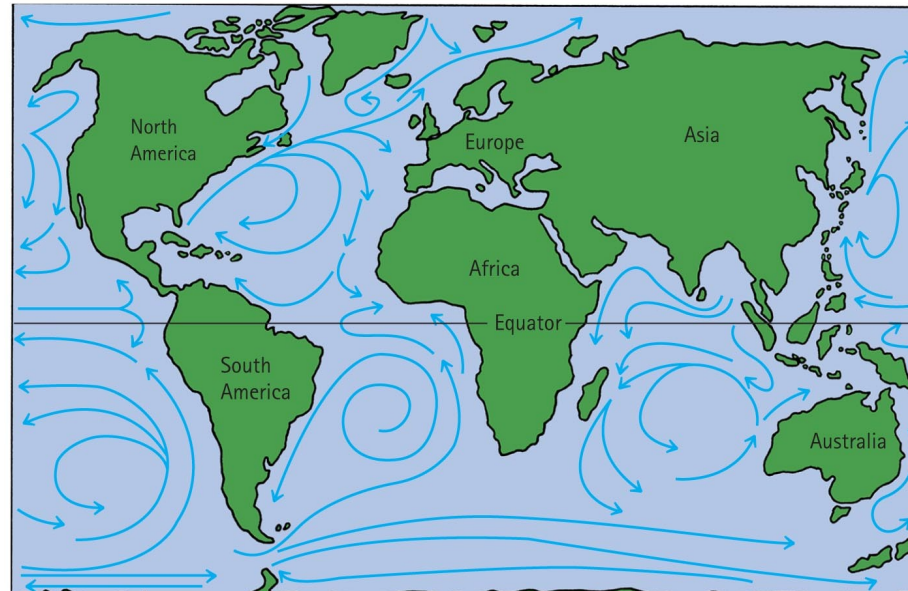
- Equal masses of different materials required different quantities of heat to change their temperatures by a specified number of degrees.
 - 1 gram of water requires 1 calorie of energy to raise the temperature 1 degree Celsius.
 - 1 gram of iron requires $\frac{1}{8}$ as much energy for the same temperature increase. Therefore, water absorbs more heat than iron for the same change in temperature. Water has a higher specific heat.

Specific Heat Capacity, Continued-2

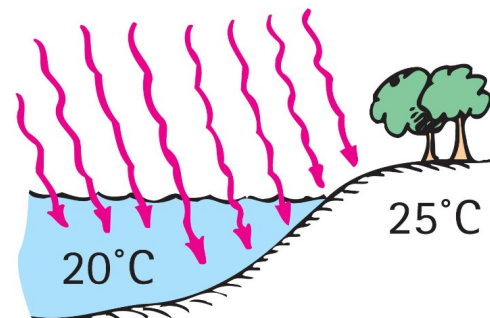
- The high specific heat capacity of water
 - has higher capacity for storing energy than almost any other substance.
 - involves various ways that energy can be absorbed.
 - increases the jiggling motion of molecules, which raises the temperature
 - increases the amount of internal vibration or rotation within the molecules, which becomes potential energy and doesn't raise temperature
 - water molecules can absorb energy without increasing translational kinetic energy

Specific Heat Capacity, Continued-3

- Specific heat affects climate
 - For Europeans, the Atlantic Ocean current carries warm water northeast from the Caribbean regions and retains much of its internal energy long enough to reach the North Atlantic Ocean. Energy released is carried by westerly winds over the European continent.



- Specific heat affects climate (continued)
 - In the United States, winds in North America are mostly westerly. On the West Coast, air moves from the Pacific Ocean to the land. In winter months, the ocean water is warmer than the air. Air blows over the warm water and then moves over the coastal regions. This produces a warm climate.
 - On the East Coast, air moves from the land to the Atlantic Ocean. Land with lower specific heat capacity gets hot in the summer and cool in the winter.



12.7 Heat and Temperature Change: Specific Heat Capacity

Example 9 A Hot Jogger

In a half-hour, a 65-kg jogger can generate $8.0 \times 10^5 \text{ J}$ of heat. This heat is removed from the body by a variety of means, including the body's own temperature-regulating mechanisms. If the heat were not removed, how much would the body temperature increase?



12.7 Heat and Temperature Change: Specific Heat Capacity

Example 9 A Hot Jogger

In a half-hour, a 65-kg jogger can generate $8.0 \times 10^5 \text{ J}$ of heat. This heat is removed from the body by a variety of means, including the body's own temperature-regulating mechanisms. If the heat were not removed, how much would the body temperature increase?

$$Q = mc \Delta T$$


$$\Delta T = \frac{Q}{mc} = \frac{8.0 \times 10^5 \text{ J}}{(65 \text{ kg})[3500 \text{ J}/(\text{kg} \cdot \text{C}^\circ)]} = 3.5 \text{ C}^\circ$$

Power Failure!

- House temperature drops to 5°C (about 41°F).
- How long to warm to room temperature (20°C)?
 - Heating system power: 30 kW
 - House's heat capacity: 25 MJ/ $^{\circ}\text{C}$



In the Kitchen: A Stove Burner

- Burner output: 1500 W (assume all goes into water)
- How long to bring 2 kg ($\sim$ 2 quarts) of water from 10°C to boiling?



In the Kitchen: A Stove Burner

- Burner output: 1500 W (assume all goes into water)
- How long to bring 2 kg (~ 2 quarts) of water from 10°C to boiling?

Solution:

It takes 4180 joules to increase the temperature of 1 kg of water by 1°C

So it takes 4,180,000 J to increase the temperature of 1 kg by 10°C

Or $2 \times 4,180,000$ joules for 2 kg.

The burner gives out 1500 J /s

So time in seconds to increase by 10°C the 2 kg is $2 \times 4,180,000 / 1500$

Then divide by 3,600 to get the hours.

Coming to Equilibrium



Insulation

Energy conservation: $Q_{1 \rightarrow 2} = -Q_{2 \rightarrow 1}$

Math convention:

Outflow negative, inflow positive: $Q_1 + Q_2 = 0$

In terms of specific heat: $m_1 c_1 \Delta T_1 + m_2 c_2 \Delta T_2 = 0$

At the Blacksmith's

- Plunge 1.2-kg iron horseshoe at $T_h = 650^\circ\text{C}$ (1200°F) into 6 gallons (23 kg) of water at $T_w = 20^\circ\text{C}$.

Find the final temperature. $M_1 c_1 (T_f - T_h) = -M_2 c_2 (T_f - T_w)$

Solve for T_f . You see how hard to increase the temp of water. $C_1=447$ and $c_2=4180$ (answer is 23.5°C)

At the Blacksmith's

- Plunge 1.2-kg iron horseshoe at $T_h = 650^\circ\text{C}$ (1200°F) into 6 gallons (23 kg) of water at $T_w = 20^\circ\text{C}$.
- Final temperature of water and horseshoe? (Neglect heat loss to surroundings.)

Energy conservation: $m_h c_h \Delta T_h + m_w c_w \Delta T_w = 0$

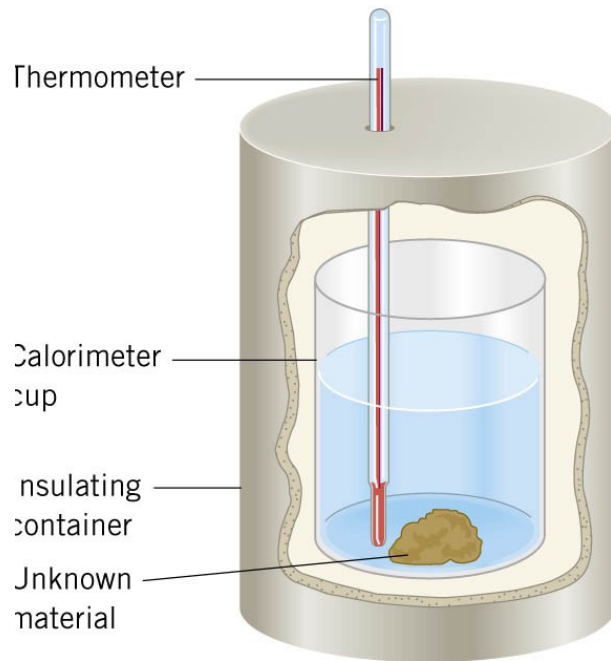
In terms of T : $m_h c_h (T - T_h) + m_w c_w (T - T_w) = 0$

Solve:
$$T = \frac{m_h c_h T_h + m_w c_w T_w}{m_h c_h + m_w c_w}$$

Do the numbers: $T_h = 650^\circ\text{C}$, $T_w = 20^\circ\text{C}$, $m_h = 1.2\text{ kg}$,
 $m_w = 23\text{ kg}$, $c_h = 447\text{ J/kg}\cdot^\circ\text{C}$, $c_w = 4184\text{ J/kg}\cdot^\circ\text{C}$

12.7 Heat and Temperature Change: Specific Heat Capacity

CALORIMETRY



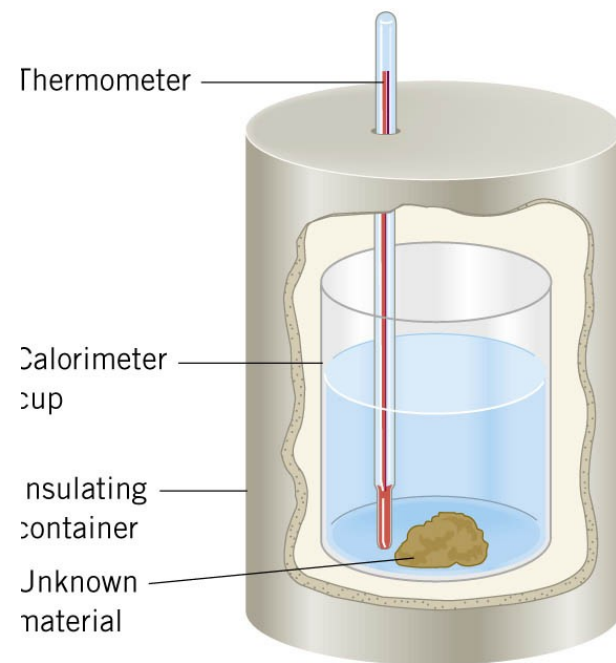
If there is no heat loss to the surroundings, the heat lost by the hotter object equals the heat gained by the cooler ones.

12.7 Heat and Temperature Change: Specific Heat Capacity

Example 12 Measuring the Specific Heat Capacity

The calorimeter is made of 0.15 kg of aluminum and contains 0.20 kg of water. Initially, the water and cup have the same temperature of 18.0°C. A 0.040 kg mass of unknown material is heated to a temperature of 97.0°C and then added to the water.

After thermal equilibrium is reached, the temperature of the water, the cup, and the material is 22.0°C. Ignoring the small amount of heat gained by the thermometer, find the specific heat capacity of the unknown material.



$$(mc \Delta T)_{\text{Al}} + (mc \Delta T)_{\text{water}} = (mc \Delta T)_{\text{unknown}}$$

12.7 Heat and Temperature Change: Specific Heat Capacity

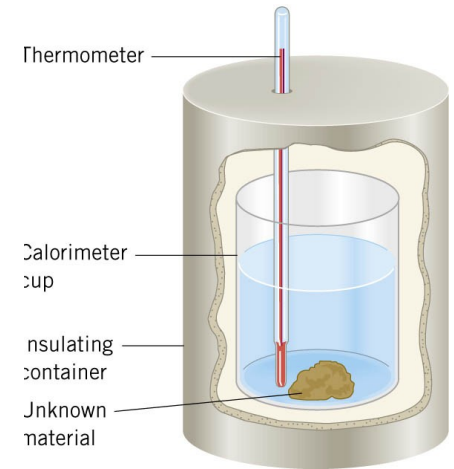
$$(mc \Delta T)_{\text{Al}} + (mc \Delta T)_{\text{water}} = (mc \Delta T)_{\text{unknown}}$$



$$c_{\text{unknown}} = \frac{(mc \Delta T)_{\text{Al}} + (mc \Delta T)_{\text{water}}}{(m \Delta T)_{\text{unknown}}}$$

$$i \frac{[9.00 \times 10^2 \text{ J}/(\text{kg} \cdot \text{C}^\circ)](0.15 \text{ kg})(4.0 \text{ C}^\circ) + [4186 \text{ J}/(\text{kg} \cdot \text{C}^\circ)](0.20 \text{ kg})(4.0 \text{ C}^\circ)}{(0.040 \text{ kg})(75.0 \text{ C}^\circ)}$$

$$i 1300 \text{ J}/(\text{kg} \cdot \text{C}^\circ)$$



Heat: energy in transit because of a temperature difference

Absolute temperature scales: zero is at absolute zero

Internal energy: the sum of kinetic and potential energies in an object

Heat capacity and **specific heat** measure energy needed for a given temperature rise

Substances at different initial temperatures come to an equilibrium determined by energy conservation